$$\lim_{x\to+\infty} \sqrt{x^2+1} - 2x =$$

$$\lim_{x\to+\infty} \sqrt{x^2} - 2x = \lim_{x\to+\infty} |x| - 2x$$

$$\lim_{x\to+\infty} x - 2x = \lim_{x\to+\infty} -x = -\infty$$

$$\lim_{x\to+\infty} \frac{\left(\sqrt{x^2+1} - 2x\right)\left(\sqrt{x^2+1} + 2x\right)}{\sqrt{x^2+1} + 2x} =$$

$$= \lim_{x\to+\infty} \frac{x^2+1-4x^2}{\sqrt{x^2+1} + 2x} = \lim_{x\to+\infty} \frac{-3x^2+1}{\sqrt{x^2+1} + 2x}$$

$$\lim_{x\to+\infty} \frac{-3x^2}{\sqrt{x^2} + 2x} = \lim_{x\to+\infty} \frac{-3x^2}{3x} = -\infty$$

$$\lim_{x\to-\infty} \frac{2x^2 + |x^2-4|}{2x^2 - |x^2-4|}$$

$$x^2-4 > 0 \qquad x < -2 \vee x > 2$$

$$\lim_{x\to-\infty} \frac{2x^2 + x^2 - 4}{2x^2 - x^2 + 4} = \lim_{x\to-\infty} \frac{3x^2-4}{x^2+4} = 3$$

Determinare il valore di $a, b \in \mathbb{R}$ in modo che

$$\lim_{x \to +\infty} \frac{x^2+x+1}{2x-1} - ax - b = -1$$

$$\lim_{x \to +\infty} \frac{x^2+x+1-2ax^2-2bx+ax+b}{2x-1} =$$

$$\lim_{x \to +\infty} \frac{x^2(1-2a)+x(1-2b+a)+1+b}{2x-1}$$

$$1-2a=0 \Rightarrow a=\frac{1}{2}$$

$$\lim_{x \to +\infty} \frac{x\left(\frac{3}{2}-2b\right)+1+b}{2x-1} = -1$$

$$\frac{\frac{3}{2}-2b}{2} = -1 \qquad \frac{3}{4}-b=-1$$

$$b = \frac{7}{4}$$

$$\lim_{x \to -\infty} \frac{\sqrt[3]{x^4+x-1}}{x+\sqrt[3]{x^2}} = \lim_{x \to -\infty} \frac{\sqrt[3]{x^4}}{x+x^{2/3}}$$

$$\lim_{x \to -\infty} \frac{x^{4/3}}{x} = \lim_{x \to -\infty} x^{1/3} = -\infty$$

$$\lim_{x\to-\infty} \sqrt{x^2-4x+3}+x$$

$$\lim_{x\to-\infty} \sqrt{x^2}+x = \lim_{x\to-\infty} |x|+x =$$

$$\lim_{x\to-\infty} -x+x \qquad \text{NON VA BENE}$$

$$\lim_{x\to-\infty} \frac{\sqrt{x^2-4x+3}+x}{\sqrt{x^2-4x+3}-x}\left(\sqrt{x^2-4x+3}-x\right)$$

$$\lim_{x\to-\infty} \frac{x^2-4x+3-x^2}{\sqrt{x^2-4x+3}-x}$$

$$\lim_{x\to-\infty} \frac{-4x}{-x-x} = \lim_{x\to-\infty} \frac{-4x}{-2x} = 2$$

$$\lim_{x\to+\infty} \frac{2e^x+5}{6-4e^x} = \lim_{x\to+\infty} \frac{2e^x}{-4e^x} = -\frac{1}{2}$$

$$\lim_{x\to-\infty} \frac{x^3+2^{-x}}{\sin x+x+3^x} = \lim_{x\to-\infty} \frac{2^{-x}}{x} = -\infty$$



$$\lim_{x\to-\infty} \frac{2e^x - x^3}{x^2 + \log|x| + 3e^x} =$$

$$\lim_{x\to-\infty} \frac{-x^3}{x^2} = +\infty$$

$$\lim_{x\to+\infty} \frac{e^{-x^2+3x-2} + 4x - 1}{\sqrt{3x} + 5}$$

$$\lim_{x\to+\infty} \frac{4x}{\sqrt{3x}} = +\infty$$

$$\sqrt{9^x} = \sqrt{(3^x)^2} = 3^x$$

$$\lim_{x\to+\infty} \frac{10^{\sqrt{x}} + x^6\cos x + \sqrt{9^x+4}}{2^{2x+1} + x^{20} + e^x}$$

$$\lim_{x\to+\infty} \frac{10^{\sqrt{x}} + 3^x}{2^x \cdot 2 + e^x} = \lim_{x\to+\infty} \frac{10^{\sqrt{x}} + 3^x}{4^x \cdot 2 + e^x}$$

$$\lim_{x\to+\infty} \frac{10^{\sqrt{x}}}{3^x} = \lim_{x\to+\infty} \frac{e^{\log 10^{\sqrt{x}}}}{e^{\log 3^x}}$$

$$\lim_{x\to+\infty} \frac{e^{\sqrt{x}\log 10}}{e^{x\log 3}} = 0$$

$$\lim_{x\to+\infty} \frac{3^x}{2\cdot 4^x} = \lim_{x\to+\infty} \frac{1}{2}\cdot\left(\frac{3}{4}\right)^x = 0$$

$$\lim_{x\to 0} \frac{\sin x}{x} = 1 \qquad \sin x \sim x \quad x\to 0$$

$$\lim_{x\to 0} \frac{1-\cos x}{\frac{1}{2}x^2} = \frac{1}{2} \qquad 1\cdot 1-\cos x \sim \frac{1}{2}x^2 \quad x\to 0$$

$$\lim_{x\to 0} \frac{\log(1+x)}{x} = 1 \qquad \cdot \log(1+x)\sim x \quad x\to 0$$

$$\lim_{x\to 0} \frac{e^x-1}{x} = 1 \qquad \cdot e^x-1 \sim x \quad x\to 0$$

$$\lim_{x\to 0} \frac{(1+x)^\alpha-1}{\alpha x} = 1 \qquad \cdot (1+x)^\alpha-1 \sim \alpha x \quad x\to 0$$

$$\cdot \tan x \sim x \quad x\to 0$$

$$\lim_{x\to 0} \frac{(1-\cos\sqrt{3}x)\cdot \sin^2 x}{x^3 \tan 2x}$$

$$\lim_{x\to 0} \frac{\frac{1}{2}(\sqrt{3}x)^2\cdot(x)^2}{x^3\cdot(2x)} =$$

$$\lim_{x\to 0} \frac{\frac{3}{2}x^4}{2x^4} = \frac{3}{4}$$

$$\lim_{x\to\pm\infty} |x| \log\left(\frac{x^2+x+1}{x^2+2}\right)$$

$\log(1+f(x)) \sim f(x)$ $\quad f(x)\to 0$

$\log(f(x)) \sim f(x)-1$ $\quad f(x)\to 1$

perché $\log(f(x)) = \log(f(x)+1-1)$

$\log\left(1+\underbrace{(f(x)-1)}_{\to 0}\right) \sim f(x)-1$ $\quad f(x)\to 1$

$$\lim_{x\to\pm\infty} |x| \log\left(\frac{x^2+x+1}{x^2+2} - 1 + 1\right)$$

$$\lim_{x\to\pm\infty} |x| \log\left(1 + \frac{x^2+x+1-x^2-2}{x^2+2}\right)$$

$$\lim_{x\to\pm\infty} |x| \log\left(1 + \frac{x-1}{x^2+2}\right)$$

$\log(1+f(x)) \sim f(x)$ $\quad f(x)\to 0$

$$\lim_{x\to\pm\infty} |x| \cdot \frac{x-1}{x^2+2}$$

$$= \begin{cases} \lim\limits_{x\to+\infty} \dfrac{x^2-x}{x^2+2} = 1 \\ \lim\limits_{x\to-\infty} \dfrac{-x^2+x}{x^2+2} = -1 \end{cases}$$

OPPURE:

$$\lim_{x\to\pm\infty} |x| \log\left(\frac{x^2+x+1}{x^2+2}\right)$$

$\frac{x^2+x+1}{x^2+2} \to 1$

$$\lim_{x\to\pm\infty} |x| \left(\frac{x^2+x+1}{x^2+2} - 1\right)$$

$$\lim_{x\to\pm\infty} |x| \left(\frac{x^2+x+1-x^2-2}{x^2+2}\right)$$

$$\lim_{x\to\pm\infty} |x| \cdot \frac{x-1}{x^2+2} = \begin{cases} 1 & x\to+\infty \\ -1 & x\to-\infty \end{cases}$$

$$\lim_{x\to\pm\infty} \left(\frac{x^2-2x}{x^2+1}\right)^{3x}$$

$$\lim_{x\to\pm\infty} e^{\log\left(\frac{x^2-2x}{x^2+1}\right)^{3x}}$$

$\log(f(x)) \sim f(x)-1$ se $f(x)\to 1$

$$\lim_{x\to\pm\infty} e^{3x \log\left(\frac{x^2-2x}{x^2+1}\right)}$$

($f(x)$)

$$\lim_{x\to\pm\infty} e^{3x\left(\frac{x^2-2x}{x^2+1} - 1\right)}$$

($f(x)$)

$$\lim_{x\to\pm\infty} e^{3x\left(\frac{x^2-2x-x^2-1}{x^2+1}\right)}$$

$$\lim_{x\to\pm\infty} e^{3x\cdot\left(\frac{-2x-1}{x^2+1}\right)} = \lim_{x\to\pm\infty} e^{\frac{-6x^2-3x}{x^2+1}} = e^{-6}$$

$$\lim_{x\to\pm\infty} \sqrt[3]{x^3+x^2}-x$$

$$\lim_{x\to\pm\infty} \sqrt[3]{x^3\left(1+\frac{1}{x}\right)}-x$$

$$\lim_{x\to\pm\infty} x\sqrt[3]{1+\frac{1}{x}}-x$$

$$\lim_{x\to\pm\infty} x\left(\sqrt[3]{1+\frac{1}{x}}-1\right) \qquad \alpha=\frac{1}{3}$$

$f(x)\to 0$

$$(1+f(x))^{\alpha}-1 \sim \alpha f(x) \quad f(x)\to 0$$

$$\lim_{x\to\pm\infty} x\cdot\frac{1}{3}\cdot\frac{1}{x}=\frac{1}{3}$$

$$\lim_{x\to+\infty} x\left(e^{\frac{x+5}{3x-2}} - \sqrt[3]{e}\right)$$

$$\lim_{x\to+\infty} \sqrt[3]{e}\,x\left(e^{\frac{x+5}{3x-2}-\frac{1}{3}} - 1\right)$$

$$\lim_{x\to+\infty} \sqrt[3]{e}\,x\left(e^{\frac{3x+15-3x+2}{3(3x-2)}} - 1\right)$$

$$\lim_{x\to+\infty} \sqrt[3]{e}\,x\left(e^{\frac{17}{9x-6}} - 1\right) \qquad \frac{17}{9x-6} \to 0$$

$$\lim_{x\to+\infty} \sqrt[3]{e}\,x \cdot \frac{17}{9x-6} \qquad e^{f(x)} - 1 \sim f(x),\ f(x)\to 0$$

$$\lim_{x\to+\infty} \frac{17\sqrt[3]{e}\,x}{9x-6} = \frac{17\sqrt[3]{e}}{9}$$

$$\lim_{x\to-\infty} \left(2^x + \log|x| + e^{1/x}\right)\left(\sqrt{x^2-1} - x\right)$$

$$\lim_{x\to-\infty} \left(2^x + \log|x| + e^{1/x}\right)\left[\sqrt{x^2\left(1-\frac{1}{x^2}\right)} - x\right]$$

$$\lim_{x\to-\infty} \left(2^x + \log|x| + e^{1/x}\right)\left[|x|\sqrt{1-\frac{1}{x^2}} - x\right]$$

$$\lim_{x\to-\infty} \left(2^x + \log|x| + e^{1/x}\right)\left[-x\sqrt{1-\frac{1}{x^2}} - x\right]$$

$$\lim_{x\to-\infty} \left(2^x + \log|x| + e^{1/x}\right)\left[-x\left(\sqrt{1-\frac{1}{x^2}} + 1\right)\right]$$

$$\lim_{x\to-\infty}\left(2^x+\log|x|+e^{1/x}\right)\left[-x\left(\sqrt{1+\left(-\frac{1}{x^2}\right)}-1+2\right)\right]$$

$$\lim_{x\to-\infty}\left(2^x+\log|x|+e^{1/x}\right)\left[-x\left(\frac{1}{2}\left(-\frac{1}{x^2}\right)+2\right)\right]$$

$(1+f(x))^\alpha-1 \sim \alpha f(x) \quad f(x)\to 0$

$\sqrt{1+\left(-\frac{1}{x^2}\right)}-1$

$\hookrightarrow f(x)\to 0 \quad \alpha=\frac{1}{2}$

$$\lim_{x\to-\infty}\left(2^x+\log|x|+e^{1/x}\right)\left(\frac{1}{2x}-2x\right)$$

$$\lim_{x\to-\infty}\log|x|\left(\frac{1-4x^2}{2x}\right)$$

$$\lim_{x\to-\infty}\log|x|\left(\frac{-2\cancel{4}x^{\cancel{2}}}{\cancel{2}x}\right)$$

$$\lim_{x\to0^+}\frac{\log^2\left(1+\sqrt{3}x^2\right)}{\left(1-\cos^2 5x\right)\left(e^{2x}-1\right)^2} \qquad e^x-1\sim x \ \ x\to0$$

$$\lim_{x\to0^+}\frac{\left(\sqrt{3}x^2\right)^2}{(1-\cos 5x)(1+\cos 5x)(2x)^2}$$

$$\lim_{x\to0^+}\frac{3x^4}{\frac{1}{2}\cdot(5x)^2(1+\cos 5x)4x^2}$$

$$\lim_{x\to 0^+} \frac{3x^4}{50x^4(1+\cos 5x)} = \frac{3}{100}$$

$$\lim_{x\to 1^+} \frac{e^{x-1}-1}{1-\cos(1-x)} \qquad x-1=t \quad x=t+1$$

$$\lim_{t\to 0^+} \frac{e^t-1}{1-\cos(-t)} = \lim_{t\to 0^+} \frac{e^t-1}{1-\cos t}$$

$$= \lim_{t\to 0^+} \frac{t}{\frac{1}{2}t^2} = \lim_{t\to 0^+} \frac{2}{t} = +\infty$$

$$\lim_{x\to -1} \frac{1-\cos(x^2-1)}{e^{x+1}-1} \qquad t=x+1 \quad x=t-1$$

$$\lim_{x\to -1} \frac{1-\cos[(x-1)(x+1)]}{e^{x+1}-1}$$

$$\lim_{t\to 0} \frac{1-\cos[(t-1-1)\cdot t]}{e^t-1}$$

$$\lim_{t\to 0} \frac{1-\cos[t(t-2)]}{e^t-1}$$

$$\lim_{t\to 0} \frac{\frac{1}{2}(t(t-2))^2}{t} = \lim_{t\to 0} \frac{\frac{1}{2}t^2(t-2)^2}{t} \to 0$$

$$\lim_{x\to\pi} \frac{1}{\sin^2 x} - \frac{1}{1+\cos x} \qquad x-\pi = t \quad x = t+\pi$$

$$\lim_{t\to 0} \frac{1}{(\sin(\pi+t))^2} - \frac{1}{1+\cos(\pi+t)}$$

$$\lim_{t\to 0} \frac{1}{(-\sin t)^2} - \frac{1}{1-\cos t}$$

$$\lim_{t\to 0} \frac{1}{t^2} - \frac{1}{\frac{1}{2}t^2}$$

$$\lim_{t\to 0} \frac{1}{t^2} - \frac{2}{t^2} = \lim_{t\to 0} -\frac{1}{t^2} = -\infty$$

$$\lim_{n\to+\infty} \left[(n^3+5n-1)^{1/3} - n\right] \log(2+e^n)$$

$$\lim_{n\to+\infty} \left[\left(n^3\left(1+\frac{5}{n^2}-\frac{1}{n^3}\right)\right)^{1/3} - n\right] \log(2+e^n)$$

$$\lim_{n\to+\infty} \left[n\left(1+\frac{5}{n^2}-\frac{1}{n^3}\right)^{1/3} - n\right] \log(2+e^n)$$

$$\lim_{n\to+\infty} n\left[\left(1+\underbrace{\frac{5}{n^2}-\frac{1}{n^3}}_{\to 0}\right)^{1/3}-1\right]\log(2+e^n)$$

$$\lim_{n\to+\infty} n\cdot\frac{1}{3}\left(\frac{5}{n^2}-\frac{1}{n^3}\right)\log(2+e^n)$$

$$\lim_{n\to+\infty} \frac{1}{3}\left(\frac{5}{n}-\frac{1}{n^2}\right)\log e^n$$

$$\lim_{n\to+\infty} \frac{1}{3}\left(\frac{5}{n}-\frac{1}{n^2}\right)\cdot n = \frac{5}{3}$$